

Employee Workforce and Salary Analysis Using SQL

Portfolio Project

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Tools Used

SQL

DB Browser for SQLite

Database : Employee & Department Database

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Project Overview

This project presents a workforce analysis conducted using SQL on an employee dataset. The objective was to answer key business and human resource questions by analyzing employee records, department information, salaries, and years of service. SQL queries were used to generate insights that support informed decision making within the organization.

The analysis focuses on four main areas: workforce overview, salary analysis, HR insights and business recommendations. Each section includes SQL queries, the corresponding results and interpretations that demonstrate practical data analysis skills. The project concludes with recommendations based on the findings, showing how data can be used to improve workforce planning and human resource management.

Report 1: Workforce Overview

Introduction

This section provides an overview of the company's workforce by examining the total number of employees, the number of departments and the distribution of employees across departments. The analysis helps to understand the organization's structure and identify departments with the highest and the lowest staffing levels.

1.1 Question 1: How many employees does the company have?

SQL Query

```
SELECT COUNT(*) AS total_employees  
FROM employees ;
```

Query Result

	total_employees
1	15

Interpretation

The query counts the total number of employees in the organization. Based on the query results, the company has 15 employees.

1.2 Question 2: How many departments does the company have?

The following SQL query counts the total number of departments in the company.

SQL Query

```
SELECT COUNT(*) AS total_departments  
FROM departments ;
```

Query Result

	total_departments
1	5

Interpretation

The query counts the total number of departments in the company. Based on the query results, the organization has 5 departments.

1.3 Question 3: Which departments have the highest and lowest number of employees?

The following SQL query identifies the number of employees in each department and sorts the departments from the highest to the lowest employee count.

SQL Query

```
SELECT COUNT(e.employee_id) AS total_employees,  
       d.department_name  
FROM departments d  
JOIN employees e  
ON e.department_id = d.department_id  
GROUP BY d.department_name  
ORDER BY COUNT(e.employee_id) DESC;
```

Query Result

	total_employees	department_name
1	4	Finance
2	3	Sales
3	3	Marketing
4	3	IT
5	2	HR

Interpretation

The query displays the number of employees in each department and ranks the departments from the highest to the lowest employee count. The results indicate that the Finance department has the highest number of employees, while the HR department has the lowest. These findings provide insight into the organization’s workforce distribution and can support staffing and resource planning.

Report 2: Salary Analysis

Introduction

This section analyses employee salaries across the organisation. The analysis examines the average salary for each department, identifies the department with the highest average salary, and highlights the top three highest-paid employees. These insights help evaluate salary distribution and support informed compensation and workforce planning decisions.

2.1 Question 4: What is the average salary in each department?

SQL Query

```
SELECT AVG(e.salary) AS avg_salary,  
       d.department_name  
FROM employees e  
JOIN departments d  
ON e.department_id = d.department_id  
GROUP BY d.department_name ;
```

Query Result

	avg_salary	department_name
1	75000.0	Finance
2	57000.0	HR
3	72000.0	IT
4	58333.3333333333	Marketing
5	48666.6666666667	Sales

Interpretation

This query shows the average salary for each department. It helps show how compensation varies across different departments in the organisation.

2.2 Question 5: Which department has the highest average salary?

SQL Query

```
SELECT AVG(e.salary) AS avg_salary,  
       d.department_name  
FROM employees e  
JOIN departments d  
ON e.department_id = d.department_id  
GROUP BY d.department_name  
ORDER BY avg_salary DESC  
LIMIT 1;
```

Query Result

	avg_salary	department_name
1	75000.0	Finance

Interpretation

This query shows the department with the highest average salary, indicating which department is most highly compensated in the company.

2.3 Question 6: Top 3 highest-paid employees in the company

SQL Query

```
SELECT e.name,  
       e.salary,  
       d.department_name  
FROM employees e  
JOIN departments d  
ON e.department_id = d.department_id  
ORDER BY e.salary DESC  
LIMIT 3;
```

Query Result

	name	salary	department_name
1	Olivia	85000	Finance
2	Esther	80000	Finance
3	Grace	76000	IT

Interpretation

This query identifies the top three highest-paid employees in the company, showing their names, departments, and salaries. It helps highlight top earners and salary distribution at the upper level.

Report 3: HR Insights

Introduction

This section focuses on deeper human resource insights, including employees earning above the company average, departments performing above average in salary distribution, and employee tenure analysis. These insights help evaluate fairness, performance distribution and workforce stability within the organization.

3.1 Question 7: Which employees earn above the company average salary?

SQL Query

```
SELECT e.name,  
       e.salary  
FROM employees e  
WHERE salary > (  
    SELECT AVG(salary)  
    FROM employees  
);
```

Query Result

	name	salary
1	Alice	65000
2	Brian	72000
3	Esther	80000
4	Grace	76000
5	James	70000
6	Kevin	68000

Interpretation

This query identifies employees whose salaries are above the company average. It helps highlight top earners and show how salaries are distributed across the organization.

3.2 Question 8: Which departments have an average salary above the company average salary?

SQL Query

```
SELECT d.department_name,  
       AVG(e.salary) AS dept_avg_salary  
FROM employees e  
JOIN departments d  
ON e.department_id = d.department_id  
GROUP BY d.department_name  
HAVING AVG(e.salary)>(  
       SELECT AVG(salary)  
       FROM employees  
);
```

Query Result

	department_name	dept_avg_salary
1	Finance	75000.0
2	IT	72000.0

Interpretation

This query compares departmental average salaries with the overall company average. Departments above the company average indicate higher compensation levels relative to the organisation.

3.3 Question 9: Which employee has worked at the company the longest?

SQL Query

```
SELECT e.name,  
       e.hire_date  
FROM employees e  
ORDER BY e.hire_date ASC  
LIMIT 1;
```

Query Result

	name	hire_date
1	Olivia	2017-01-20

Interpretation

This query identifies the employee with the longest tenure by sorting hire dates from the earliest to the latest date. It highlights workforce stability and long-term retention.

Key Findings Summary

- The company has **15 employees** across **5 departments**.
- The **Finance** department has the highest number of employees (**4 employees**).
- The **Finance** department has the highest average salary (**KES 75,000**).
- **Alice, Brian, Esther, Grace, James, Kevin and Olivia** earn above the company average salary.
- **Olivia** is the longest-serving employee, having joined the company on **20 January 2017**.

Report 4: Final Business Recommendations

This section presents business recommendations based on the findings from the workforce overview, salary analysis, and HR insights. The recommendations are intended to support informed decision-making regarding workforce planning, compensation and employee retention.

4.1 Recommendation 1: Review Workforce Distribution

The workforce analysis showed that the company has 15 employees distributed among 5 departments. The Finance department has the highest number of employees(4), while the HR department has the lowest (2). Management should periodically review staffing levels across departments to ensure that resources are distributed effectively and that departments with fewer employees have sufficient capacity to meet their responsibilities.

4.2 Recommendation 2: Review Salary Structure

The salary analysis showed that the Finance department had the highest average salary of KES 75,000 ,while the highest-paid employees were Olivia (KES 85,000) ,Esther (KES 80,000), and Grace (KES 76,000). Management should regularly review the company's salary structure to ensure that the employee compensation remains fair, competitive, and aligned with the job roles and responsibilities across all departments.

4.3 Recommendation 3:Recognize and Retain High-Performing Employees

The HR analysis identified Alice, Brian, Esther, Grace, James, Kevin and Olivia as employees earning above the company average salary. In addition, the Finance and IT departments had average salaries above the company average, while Olivia, who joined the company on 20 January 2017, was identified as the longest-serving employee. Management should continue investing in employee retention through career development opportunities, recognition programmes, and competitive compensation to retain experienced and high-performing employees.